

02YMECH - Homework 5

Assigned for the week of Oct 20, 2025

Questions

1. Calculate the average total mechanical energy of driven damped harmonic oscillator over one complete cycle. Calculate the total energy when the driving frequency is equal to the natural oscillation frequency.
2. Calculate the power delivered by the driving force to the damped harmonic oscillator. Show that, on average, it is equal to the power dissipated by the frictional force.
3. Express the total energy of the damped harmonic oscillator in terms of time t , quality factor Q and natural oscillation frequency ω_0 . Interpret the result.